Example 1.17

Investigate for what values of 'a' and 'b' the following system of equations

$x+y+z=6,\quad x+2y+3z=10,$
$x+2y+az=b$ have (i) no solution (ii) a unique solution (iii) an infinite number of solutions.

Solution:

The matrix equation corresponding to the given system is

$$\begin{pmatrix}1&1&1\\1&2&3\\1&2&a\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=\begin{pmatrix}6\\10\\b\end{pmatrix}$$
$$A \quad X = B$$

Augmented matrix $[A,B]$	Elementary Transformation
$\begin{pmatrix}1&1&1&6\\1&2&3&10\\1&2&a&b\end{pmatrix}$	
$\sim\begin{pmatrix}1&1&1&6\\0&1&2&4\\0&1&a-1&b-6\end{pmatrix}$	$R_2\to R_2-R_1$, $R_3\to R_3-R_1$
$\sim\begin{pmatrix}1&1&1&6\\0&1&2&4\\0&0&a-3&b-10\end{pmatrix}$	$R_3\to R_3-R_2$

Case (i) For no solution:

The system possesses no solution only when $\rho(A)\neq\rho([A,B])$ which is possible only when $a-3=0$ and $b-10\neq 0$

Hence for $a=3,\ b\neq 10$, the system possesses no solution.

Case (ii) For a unique solution:

The system possesses a unique solution only when $\rho(A)=\rho([A,B])=$ number of unknowns.

i.e when $\rho(A)=\rho([A,B])=3$

Which is possible only when $a-3\neq 0$ and b may be any real number as we can observe .

Hence for $a\neq 3$ and $b\in R$, the system possesses a unique solution.

Case (iii) For an infinite number of solutions:

The system possesses an infinite number of solutions only when

$\rho(A)=\rho([A,B])<$ number of unknowns

i,e when $\rho(A)=\rho([A,B])=2<3$ (number of unknowns) which is possible only when $a-3=0,\quad b-10=0$

Hence for a = 3, b =10, the system possesses infinite number of solutions.

Example 1.18

The total number of units produced (P) is a linear function of amount of over times in labour (in hours) (l), amount of additional machine time (m) and fixed finishing time (a)

i.e, $P = a + bl + cm$

From the data given below, find the values of constants a, b and c

Day	Production (in Units P)	Labour (in Hrs l)	Additional Machine Time (in Hrs m)
Monday	6,950	40	10
Tuesday	6,725	35	9
Wednesday	7,100	40	12

Estimate the production when overtime in labour is 50 hrs and additional machine time is 15 hrs.

Solution:

We have, $P = a + bl + cm$

Putting above values we have

$6{,}950 = a + 40b + 10c$

$6{,}725 = a + 35b + 9c$

$7{,}100 = a + 40b + 12c$

The Matrix equation corresponding to the given system is

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